201 Motion

2.1.1 Distance and displacement

Displacement is the change from initial to final position and has both magnitude and direction. Distance adds all the lengths along the actual path. Returning to the starting point gives zero displacement but usually non-zero distance. In one dimension, choose a positive direction and subtract initial position from final position; the sign indicates direction.

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2.1.2 Scalars and vectors

Scalars need magnitude only, such as mass, time, distance and energy. Vectors also need direction, such as displacement, velocity, force and momentum. Vector addition must account for directions rather than simply adding magnitudes. Signs can represent direction in one dimension; in more dimensions, resolve vectors into components.

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2.1.3 · 2.1.4 Speed and average speed

Speed describes how fast an object travels along its path. Average speed is total distance divided by total elapsed time, including stops. Speeds for separate stages can be averaged arithmetically only when those stages take equal times. Speed has no direction, unlike average velocity, which is calculated from displacement.

$$v_{\text{speed,av}} = \frac{d_{\text{total}}}{\Delta t}$$

Average speed (m s⁻¹) = total distance travelled (m) / total elapsed time (s). Scalar; include rest intervals.

Textbook sources: 201 · PDF 11, 13

2.1.5 · 2.1.6 Velocity and average velocity

Velocity describes the rate and direction of position change. Average velocity is net displacement divided by total elapsed time. A round trip can have zero average velocity but non-zero average speed. Choose a positive direction or state the direction explicitly; average velocity need not equal the instantaneous velocity during a particular stage.

$$\vec{v}_{\text{av}} = \frac{\Delta \vec{s}}{\Delta t}$$

Average velocity (m s⁻¹); Δs: resultant displacement (m); Δt: total time (s). State the direction.

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2.1.7 Acceleration and velocity change

Average acceleration is final velocity minus initial velocity, divided by elapsed time. A change in speed or direction can cause acceleration. Negative acceleration means acceleration in the chosen

negative direction, not necessarily slowing down. Use a consistent sign convention for all velocities before calculating their difference.

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2.1.8 Motion at constant velocity

Constant velocity requires both constant speed and constant direction: the object moves in a straight line with equal displacements in equal times and zero acceleration. Constant speed alone is insufficient, since uniform circular motion changes direction. Zero resultant force can maintain either rest or a non-zero constant velocity.

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2.1.9 Percentage uncertainty

Percentage uncertainty compares an absolute measurement uncertainty with the measured value, indicating relative precision. With the same absolute uncertainty, a larger measured value gives a smaller percentage uncertainty. Numerator and denominator must have matching units. Percentage deviation from a reference value instead uses that reference value as the denominator; do not confuse the two calculations.

$$\text{percentage uncertainty} = \frac{\delta x}{x} \times 100\%$$

Textbook wording: percentage error = error / measured value × 100%. Here δx is the absolute measurement uncertainty and x the measured value in the same units. Deviation from a reference value uses that reference in the denominator.

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